

Stray Animal Needs Database

CS50 SQL Final Project

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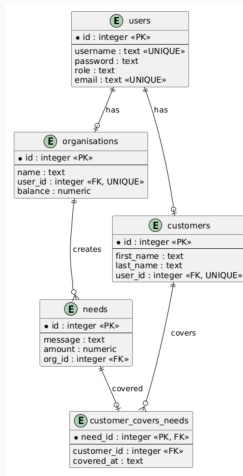
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Project Purpose

The purpose of this database is to support organisations that help stray animals, such as dogs and cats.

- Organisations create needs for stray animals.
- Needs can include food, medicine or veterinary care.
- Customers can cover the full amount of a need.
- The database tracks covered and uncovered needs.

Database Design



How It Works

1. An organisation creates a need.
2. The need contains a description and an amount.
3. A customer chooses to cover the full need.
4. The coverage is stored in `customer_covers_needs`.
5. A trigger automatically updates the organisation's balance.

Each need can only be covered once.

Database Features

Views

- `uncovered_needs`
- `covered_needs`
- `organisation_statistics`

Indexes

- `needs_org_index`
- `covers_customer_index`

Trigger

- `update_organisation_balance`

Example Queries

The database can answer questions such as:

- Which needs have not been covered?
- Which customers have covered at least one need?
- Which customers have never covered a need?
- Which uncovered needs are above the average amount?
- Which three customers have covered the largest total amount?

Stray Animal Needs Database

A simple database that connects animal welfare organisations with customers who want to help stray animals.

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